

YOGENDRA GUPTA

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PROFESSIONAL SUMMARY

Computer Science undergraduate at VIT Vellore (CGPA 8.14, Class of 2027) with hands-on experience designing and shipping backend services and full-stack applications in Python and JavaScript. Skilled in REST API design, JWT authentication, ORM-based data modeling, and Linux/Nginx deployment, with a growing focus on system design, computer vision, and applied machine learning. Known for turning ambiguous requirements into secure, maintainable, production-ready systems.

EDUCATION

VIT Vellore — B.Tech in Computer Science Engineering

2023 – 2027

CGPA: 8.14 / 10

Kendriya Vidyalaya, Indore — Class XII (CBSE)

2023

Percentage: 75.2%

Kendriya Vidyalaya, Indore — Class X (CBSE)

2021

Percentage: 81.4%

TECHNICAL SKILLS

Languages	Python, Java, C++, SQL, JavaScript
Backend	FastAPI, REST APIs, JWT Authentication & Authorization, SQLAlchemy ORM, CRUD Design
Frontend	React, Streamlit, Tkinter
AI / ML	OpenCV, Object Detection, NumPy, Pandas
Databases	SQLite, Database Design
Cloud & DevOps	AWS EC2, Nginx, Linux (Ubuntu)
Tools	Git, GitHub, Version Control
Core CS	Data Structures & Algorithms, OOP, System Design Fundamentals

PROJECTS

FastAPI Backend Service | [GitHub](#)

Engineered a production-grade REST backend that authenticates users and serves modular business logic on a live cloud deployment.

- Architected a layered FastAPI service (routers → services → data-access) to isolate auth, business logic, and persistence, reducing cross-module coupling and easing future feature additions.
- Implemented JWT-based authentication with protected-route dependencies and request validation, securing every API endpoint against unauthorized access.
- Replaced raw SQL with SQLAlchemy ORM across the data layer, eliminating duplicated query logic and closing off SQL-injection vectors.
- Deployed and hardened the service on an AWS EC2 (Ubuntu) instance behind an Nginx reverse proxy running Uvicorn, managing releases through Git/SSH for zero-downtime updates.

YOLO Playground — Object Detection App | [GitHub](#)

Built an end-to-end computer vision pipeline that turns raw images into real-time, confidence-scored detections.

- Designed a YOLO + OpenCV inference pipeline covering preprocessing, normalization, and bounding-box post-processing for real-time object detection.
- Tuned confidence-based classification thresholds to balance detection precision and recall across varied image inputs.
- Built a Streamlit interface for image upload and result visualization, removing the CLI barrier for non-technical users.

Chrome Highlight Extension | [GitHub](#)

Shipped a lightweight browser extension that captures and syncs highlighted web content without a dedicated backend.

- Built a Chrome extension with JavaScript and the Extension APIs to capture highlighted text plus source URL and timestamp metadata.
- Integrated Google Apps Script with Google Sheets as a serverless storage backend, eliminating the need for dedicated database infrastructure.
- Engineered an asynchronous capture-and-sync flow to keep the extension non-blocking and responsive during active browsing.

Inventory Management System | [GitHub](#)

Designed an extensible inventory platform with decoupled front ends, laying groundwork for a future Flask web deployment.

- Modularized core business logic away from CLI and Tkinter GUI layers, enabling multiple front ends to share a single codebase.
- Implemented full CRUD operations with input validation and structured error handling to prevent invalid state changes.
- Structured the codebase for extensibility, enabling a planned Flask-based web version without rewriting core logic.

CERTIFICATIONS

- Oracle AI Foundations Associate — Oracle
- Introduction to Data Science — Infosys

ACHIEVEMENTS & HACKATHONS

Participant — VIT Vellore Hackathon

- Contributed to rapid prototyping and frontend development, collaborating with a cross-functional team to design and present a functional prototype under time constraints.